Lab 06 Reflection

Antonio Garcia

In Lab 06, I worked with two different types of machine learning problems: regression and classification. Even though I've seen these terms before, actually building both models side by side helped me understand the difference much more clearly. Regression predicts a continuous number, while classification predicts a category. In this lab, predicting fare was a regression task, and predicting survival was a classification task. Seeing how each model behaved with the Titanic dataset made the concepts feel more real instead of just definitions.

Training the linear regression model showed me how features like Age and Pclass relate to Fare. When I looked at the coefficients, I realized they were basically the model's way of saying how much each feature pushes the fare up or down. The evaluation step also taught me why mean squared error matters. It doesn't just tell me whether the model is "right" or "wrong," but how far off the predictions are. When I calculated the square root of the MSE, it gave me a more intuitive idea of the average error in dollars. That made the metric feel practical instead of abstract.

The logistic regression model helped me understand classification better. Predicting survival is a yes-or-no problem, so accuracy made sense as the evaluation metric. It

was interesting to see how adding the Sex feature changed the model's ability to separate survivors from non-survivors. I also learned why accuracy wouldn't work for regression. If I tried to measure fare prediction with accuracy, the model would basically always be wrong because fares are continuous numbers. That helped me understand why choosing the right metric is just as important as choosing the right model.

One challenge I ran into was the code cells breaking when pasted into Colab. Some lines merged together, causing NameErrors and undefined variables. Fixing those issues forced me to slow down and make sure each part of the workflow was set up correctly. It also made me pay more attention to how the data was prepared, especially for the classification task where the Sex column had to be encoded. Once everything was fixed, the models trained smoothly, and the results made sense. Overall, this lab helped me understand the workflow of building and evaluating machine learning models. I learned how regression and classification differ, how model coefficients can be interpreted, and why evaluation metrics matter. I also got more comfortable debugging issues in Colab and making sure the data is prepared correctly before training. This lab strengthened my understanding of machine learning basics and made me feel more confident moving forward.

Cristian G Delgado

Regression and classification are different methods of machine learning that focus on different problems. Regression, for example, is used for predicting numerical values like house prices, whereas classification focuses on categorizing, where it attempts to predict of a subject being one thing over the other. The lab itself provides examples of basic ways these two methods are employed for simple problems.

Only difficulty I had was the initial formatting made it difficult to read (all code in cells being written on a single line) but once it was restructured (using Google Colab built-in AI) it was relatively straight forward. Running the code cells provided usable results. One thing that surprised me was utilizing different ways to score/measure the performance of our models. While it makes sense, I figured that it came down more to preference, not realizing that different metrics varied significantly and weren't necessarily compatible with eac other.

A question I have is whether there are other ways of evaluating a model besides accuracy and mean squared error. Additionally, can these evaluation methods be used in tandem with others? Where precision and recall is concerned, are the 2 evaluation methods described in the lab related or are they different evaluation techniques altogether?

Steven Ford

In Lab, I worked through machine learning concepts using Python and scikit-learn, building and comparing regression and classification models. Even though I've studied these topics in my coursework, implementing both models side by side made the distinction much clearer. Regression predicts a continuous numerical target, while classification sorts data into discrete categories. Working directly with the dataset brought these definitions to life and showed how the theory translates into actual code execution.

Training the models helped me understand how feature relationships drive predictions. Looking at regression coefficients showed me how specific attributes influence the final numeric output, while evaluating with mean squared error gave me a practical way to measure how far off the predictions were on average. Switching to classification made the role of evaluation metrics even more apparent. Because predicting categories requires different success criteria than predicting numbers, using accuracy provided a straightforward way to measure success for a yes-or-no outcome, reinforcing why choosing the right metric matters just as much as choosing the right algorithm.

One practical challenge I ran into involved managing my development environment and handling code formatting across different notebook cells. Resolving those minor debugging issues forced me to slow down and carefully verify how data preparation steps—like feature encoding—were structured before fitting the model. Once those environment and data preparation hurdles were cleared, the workflow ran smoothly, and the output curves and metrics aligned with expectations.

Overall, this lab strengthened my practical understanding of building, debugging, and evaluating machine learning models in a Python environment. I gained a firmer grasp of how regression and classification differ, how to interpret model outputs, and why proper data preparation is essential for clean execution. Working through these steps increased my confidence in handling machine learning workflows moving forward.

Khin Thu

Lab 6 helped me understand how the prediction goal determines which machine learning model to use. Regression predicts a continuous numerical value, while classification predicts a category. Using the Titanic dataset for both tasks made this difference clearer. Predicting a passenger's fare is a regression problem because the answer is an amount of money. Predicting whether a passenger survived is a classification problem because the answer belongs to one of two categories. Both are supervised learning tasks because the dataset contains known outcomes that the models can learn from.

The linear regression section helped me connect input features to a numerical prediction. The model uses age and passenger class to predict fare. Understanding the coefficients helped me see that training involves learning relationships between the features and the target. Each coefficient represents how the predicted fare changes when one feature increases by one unit, while the other feature stays constant. However, these relationships do not prove cause and effect. I also realized that using only two features limits the information available to the model, so its predictions may not capture all the factors that influenced ticket prices.

The logistic regression section clarified something that could easily be confusing: despite its name, logistic regression is used for classification. In this task, it uses age, passenger class, and sex to estimate survival probability and produce a class prediction. Encoding the sex column as numbers showed me why data preparation matters. A model cannot simply use every column in its original form; the information must be represented in a format the algorithm can process. Removing rows with missing ages also reminded me that cleaning decisions can affect which

passengers are included in the analysis.

Another important lesson was choosing an evaluation metric that matches the task. Mean squared error measures the squared differences between actual and predicted fares, giving larger errors more weight. Its square root expresses the error in the same units as fare, making it easier to interpret. Accuracy measures the proportion of correct survival classifications. Accuracy would not be suitable for fare prediction because a numerical prediction can be close to the actual value without matching it exactly.

The workflow helped me understand how preparation, training, prediction, and evaluation depend on one another. Separating training and test data is especially important because evaluating on examples already used for training could give an overly optimistic impression. My main takeaway is to define the target clearly before selecting a model or metric. In a future project, I would examine the available features, consider the effects of cleaning choices, and evaluate predictions on unseen data before deciding whether the model is useful.